

Liverpool FC Performance Simulation

In this project, I aimed to predict the results of the first half of the English Premier League season for Liverpool FC. I divided the remaining teams into 4 categories: Favorites to win the league, Strong teams, Mediocre teams, and lastly, Weak team. A game of football lasts for 90 minutes, so we divide the entire duration of game into 10 second intervals, and in each interval, we make a transition into some state. So, for example, if in the first 20 seconds of the game Liverpool had the ball and was attacking, then $X_0 = \text{'Possess, Attack'}$, and $X_1 = \text{'Possess, Attack'}$. I created 6 states for my model: 'Possess, Defend', 'Possess, Attack', 'Lose Ball, Defend', 'Lose Ball, Press', 'Goal Score', 'Goal Concede'. 'Possess, Defend' means that the team currently has the ball but is playing defensively, 'Possess, Attack' means the team has the ball and is attacking, 'Lose Ball, Defend' means that the team lost possession of the ball and is now in defensive state, 'Lose Ball, Press' means that the team has lost the ball but is pressing to get it back, and lastly, 'Goal Score' means a goal was scored by Liverpool, and 'Goal Concede' means Liverpool conceded a goal. Then I created 4 transition matrices for each of these categories of teams:

Note that in the matrices below, 0: 'Possess, Defend', 1: 'Possess, Attack', 2: 'Lose Ball, Defend', 3: 'Lose Ball, Press', 4: 'Goal Score', 5: 'Goal Concede'.

$$= \begin{pmatrix} \frac{1}{10} & \frac{1}{2} & \frac{199}{1000} & \frac{199}{1000} & \frac{1}{500} & 0 \\ \frac{3}{20} & \frac{2}{5} & \frac{221}{1000} & \frac{221}{1000} & \frac{1}{125} & 0 \\ \frac{1591}{8000} & \frac{1591}{8000} & \frac{479}{800} & 0 & 0 & \frac{7}{2000} \\ \frac{31}{125} & \frac{31}{125} & 0 & \frac{1}{2} & 0 & \frac{1}{250} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \end{pmatrix}$$

Figure 1: Favorite Team Transition Matrix

$$= \begin{pmatrix} \frac{3}{20} & \frac{3}{4} & \frac{1}{20} & \frac{1}{20} & 0 & 0 \\ \frac{1}{10} & \frac{13}{20} & \frac{1}{10} & \frac{7}{50} & \frac{1}{100} & 0 \\ \frac{239}{800} & \frac{239}{800} & \frac{2}{5} & 0 & 0 & \frac{1}{400} \\ \frac{279}{800} & \frac{279}{800} & 0 & \frac{3}{10} & 0 & \frac{1}{400} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \end{pmatrix}$$

Figure 2: Weak Team Transition Matrix

$$= \begin{pmatrix} \frac{11}{80} & \frac{11}{16} & \frac{349}{4000} & \frac{349}{4000} & \frac{1}{2000} & 0 \\ \frac{9}{80} & \frac{47}{80} & \frac{521}{4000} & \frac{641}{4000} & \frac{19}{2000} & 0 \\ \frac{273781}{1000000} & \frac{273781}{1000000} & \frac{56211}{125000} & 0 & 0 & \frac{11}{4000} \\ \frac{161781}{500000} & \frac{161781}{500000} & 0 & \frac{7}{20} & 0 & \frac{23}{8000} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \end{pmatrix}$$

Figure 3: Mediocre Team Transition Matrix

$$= \begin{pmatrix} \frac{1}{8} & \frac{5}{8} & \frac{249}{2000} & \frac{249}{2000} & \frac{1}{1000} & 0 \\ \frac{1}{8} & \frac{21}{40} & \frac{321}{2000} & \frac{361}{2000} & \frac{9}{1000} & 0 \\ \frac{62203}{250000} & \frac{62203}{250000} & \frac{799}{1600} & 0 & 0 & \frac{3}{1000} \\ \frac{2387}{8000} & \frac{2387}{8000} & 0 & \frac{2}{5} & 0 & \frac{13}{4000} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \end{pmatrix}$$

Figure 4: Strong Team Transition Matrix

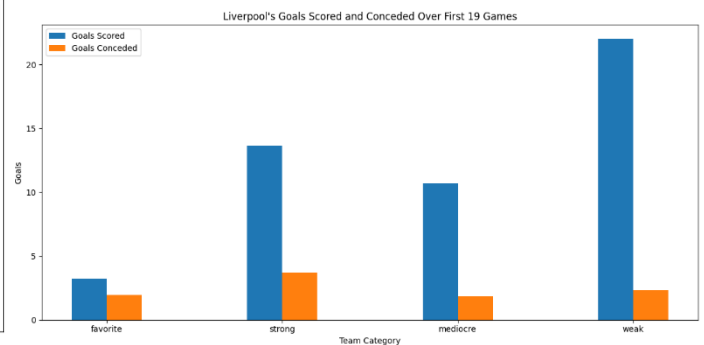
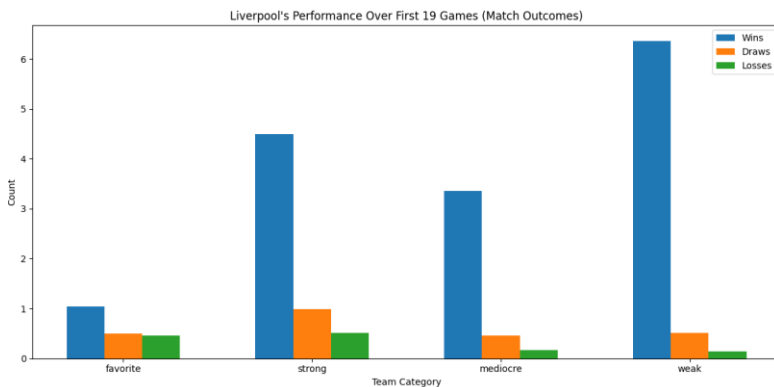
Liverpool FC Performance Simulation

1. I tried to predict the result of the first half of the season for Liverpool. Currently, the team has the following record: 11W, 2D, 1L. The following is the result of my code and the theoretical result of the first half of the season:

```
Season Statistics:  
Wins: 15.47  
Draws: 2.28  
Losses: 1.25  
Goals Scored: 50.58  
Goals Conceded: 9.45  
Clean Sheets: 11.78
```

Theoretical Season Statistics	
Wins	15.41
Draws	2.37
Losses	1.22
Goals Scored	50.45
Goals Conceded	9.61
Clean Sheets	11.67

As we can see the two results are almost identical. Additionally, the result of the code is not too far off from the ongoing real-life situation, implying that our assumptions on the transition matrix values are acceptable. The number of goals scored is a little bit excessive, as well as possession, which at the moment are at 29 goals in 14 games. Additionally, we can calculate the theoretical value of possession, which is and 67.78, just 10% off of the real value - 56.87% possession rate. Here is extra insight into the statistics of the season:



From this, we can conclude that the simulation accurately describes the performance of the team, and future games of the team may enhance this claim. Several assumptions are purely subjective and may require a more careful analysis, but such an extensive examination of the assumptions would not fit in 2 pages. In particular, the assumption on the categorization of the teams was based purely on my view of the performance of the teams, and their current league standings, but could be supplemented by the performance of the team in the last, say, 15 or 20 games.